Predicting Falcon 9 First Stage Landings

IBM Data Science Capstone | Space Y launch cost analysis

Ali | October 2026

All notebooks and code:

https://github.com/Ali2187/spacex

Executive Summary

- **Goal:** predict whether a Falcon 9 first stage will land successfully, because a reused booster is what makes SpaceX launches roughly 3x cheaper than competitors.
- **Data:** SpaceX REST API and a Wikipedia scrape, wrangled into one table of 90 Falcon 9 launches with 17 raw features and a binary landing label.
- **Analysis:** SQL, seaborn charts, a Folium launch-site map and a Plotly Dash dashboard to find what drives landing success.
- **Modelling:** four tuned classifiers (logistic regression, SVM, decision tree, KNN) compared on a held-out set of 18 launches.
- **Headline finding:** landing success rises with flight number and booster maturity. Heavy-payload Block 5 boosters now land reliably, and the early failures were concentrated in older boosters at CCAFS.

Result: Decision tree gave the best cross-validation accuracy (86.1%) and 83.3% on the 18-launch test set, tied with logistic regression and KNN. SVM scored lowest on test (77.8%).

Introduction

Business problem. SpaceX advertises Falcon 9 launches at about $62M, while competitors charge upward of $165M. Most of the saving comes from landing and reusing the first stage. A competitor (Space Y) that can predict landing success can estimate a launch's true cost before bidding.

Which factors drive landing success?

payload, orbit, site, booster block, reuse

Does success improve over time?

flight number and year trends

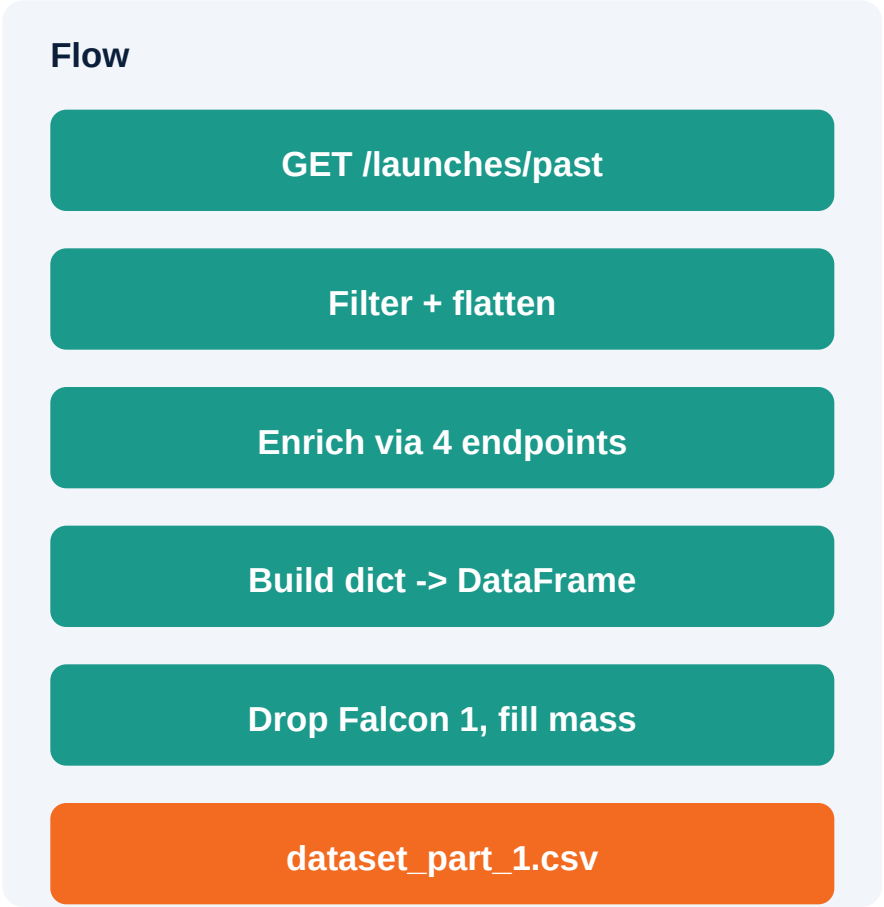
Can we predict it?

classification on 90 launches

Context from the API: Falcon 1 flew 5 times with 3 failures (40% success, $7M per launch), which shows how far the program has come.

Data Collection: SpaceX REST API

- Pulled **/v4/launches/past**, flattened with `json_normalize`.
- Kept rocket, payloads, launchpad, cores, flight number, date; dropped multi-core and multi-payload launches; limited to launches up to 2020-11-13.
- Helper functions called **/rockets, /launchpads, /payloads, /cores** to resolve IDs into booster version, site name with lat/long, payload mass, orbit, block, reuse count, serial and landing outcome.
- The live endpoint returned a **522 timeout**, so the static IBM snapshot of the same JSON was used to keep the run reproducible.
- Cleaning: removed Falcon 1, renumbered flights, filled missing **PayloadMass** with the mean; LandingPad was left null where no pad was used.



Data Collection: Web Scraping

- Source: Wikipedia *List of Falcon 9 and Falcon Heavy launches* (fixed revision `oldid=1027686922` for reproducibility).
- **BeautifulSoup** parsed every table of class `wikitable plainrowheaders collapsible`.
- Custom parsers extracted **date, time, booster version, launch site, payload, mass, orbit, customer, launch outcome and booster landing** for each numbered flight row.
- Rows were collected into a dictionary of lists and converted to a pandas DataFrame, then exported to CSV.

INSERT: Screenshot of df.head() from the web scraping notebook. Note: the saved run shows soup.title = None, so re-run it with the headers dict passed to requests.get() and confirm the tables load before screenshotting.

Data Wrangling

- Checked missing values: only **LandingPad** had gaps (28.9%); all other columns were complete.
- Counted launches by site, by orbit and by landing outcome (8 distinct outcome strings).
- Defined **bad outcomes** (False Ocean, False ASDS, False RTLS, None ASDS, None None) and created the label **Class**: 1 = landed, 0 = did not land.

Launch site	Launches
CCAFS SLC 40	55
KSC LC 39A	22
VAFB SLC 4E	13

Landing outcome	Meaning
True ASDS / RTLS / Ocean	Landed (1)
False ASDS / RTLS / Ocean	Failed (0)
None ASDS / None None	No landing (0)

Label rule: Class = 0 for False Ocean, False ASDS, False RTLS, None ASDS and None None (no landing or failed landing); Class = 1 for every successful landing. This label is the target variable for all models.

EDA and Interactive Analytics Methodology

SQL (SQLite)

- Loaded 101 launches into SPACEXTABLE
- Aggregates, sub-queries, LIKE and date filters
- Questions: sites, NASA payload, max payload, outcomes by period

Visual EDA

- seaborn catplot / bar / line
- Flight no. vs payload, site, orbit
- Class as colour to expose success patterns
- One-hot encoded Orbit, Site, Pad, Serial (80 features)

Folium map

- Marked 4 launch sites
- Marker clusters coloured green / red by outcome
- Mouse-position tool and haversine distance to coast, city, rail, highway

Plotly Dash

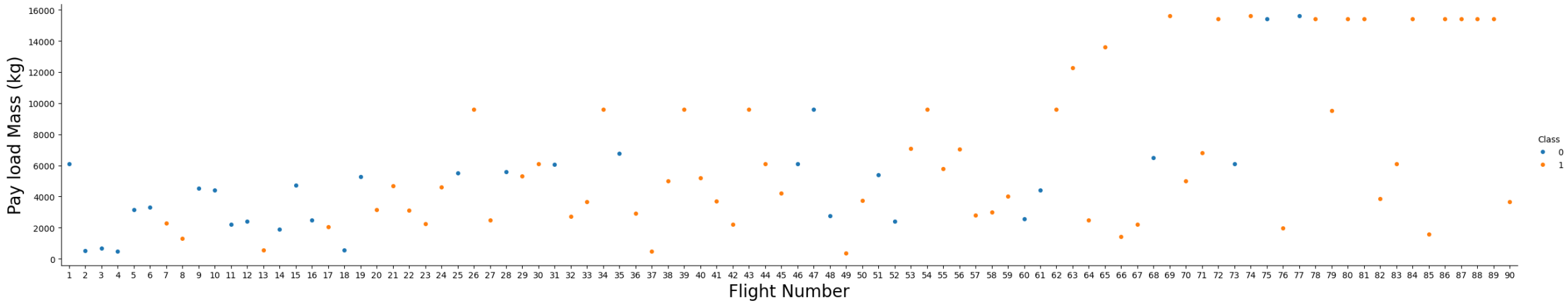
- Site dropdown + payload range slider
- Pie chart: success share by site
- Scatter: payload vs outcome by booster version

Predictive Analysis Methodology

1. Load features (dataset_part_3) and target Class
2. Standardise with StandardScaler
3. Split 80/20, random_state=2 (72 train / 18 test)
4. GridSearchCV on 4 models
5. Score on test set + confusion matrix
6. Pick best model

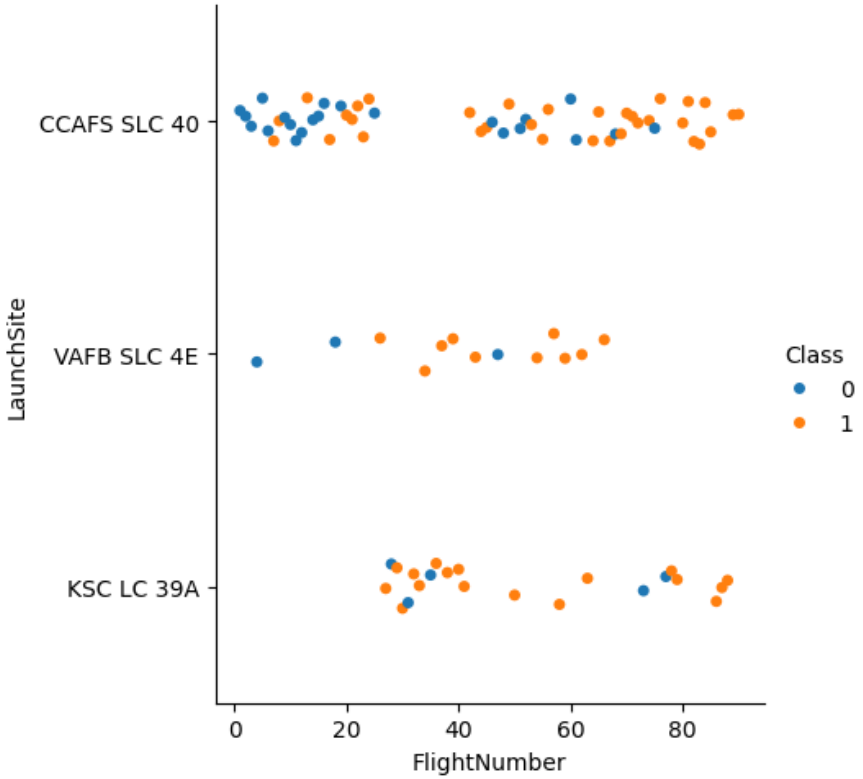
Model	Grid searched
Logistic regression	C in {0.01, 0.1, 1}, L2 penalty, lbfgs
SVM	kernel (linear, rbf, poly, sigmoid), C and gamma over logspace(-3, 3)
Decision tree	criterion, splitter, max_depth 2-18, max_features, min_samples_leaf / split
KNN	k = 1-10, algorithm (auto, ball_tree, kd_tree, brute), p = 1 or 2

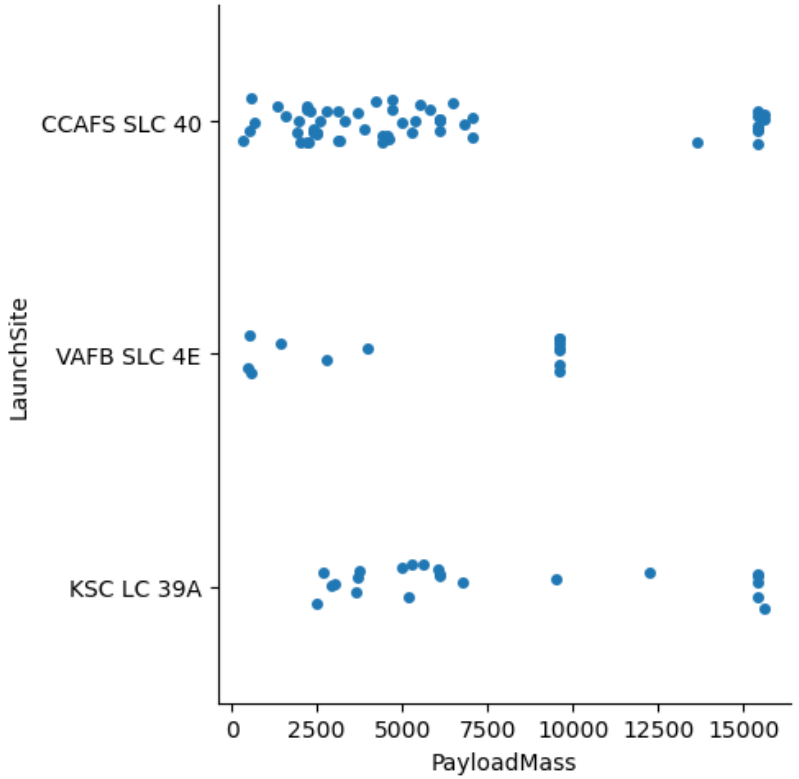
EDA with Visualization: Flight Number vs Payload



- Failures (blue) cluster in early flights; later flights are overwhelmingly successful.
- Payloads above about 13,000 kg appear only after flight 60 and mostly land successfully.

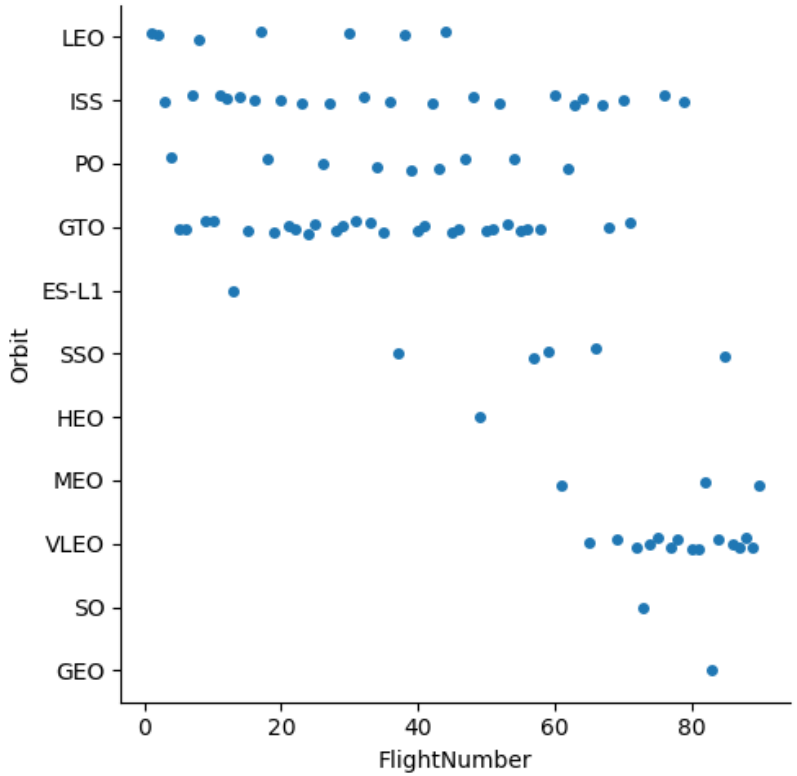
EDA with Visualization: Launch Site

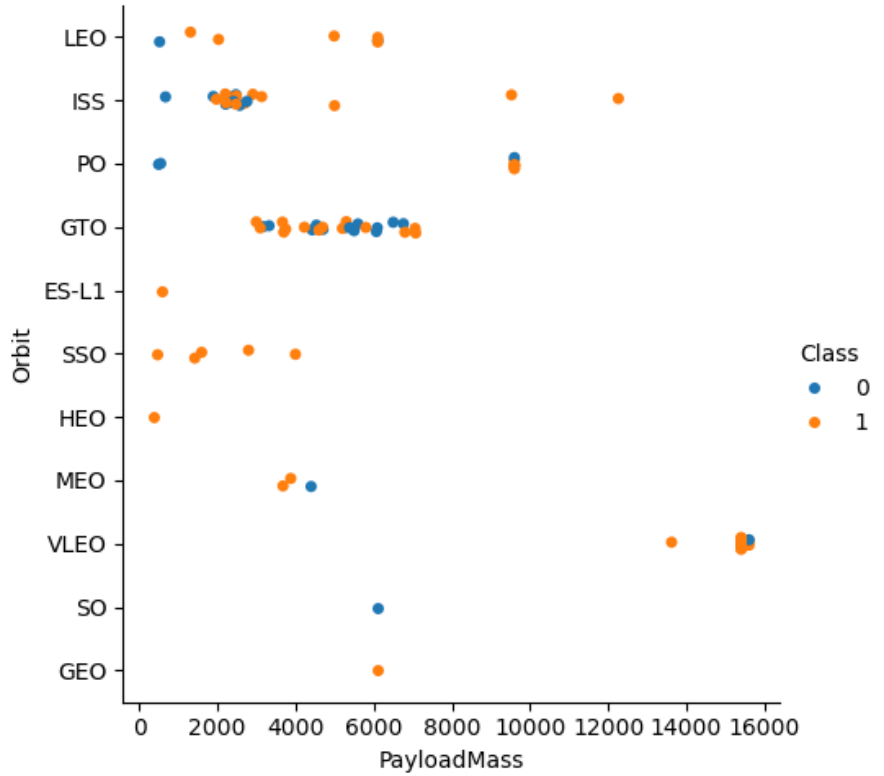




- CCAFS SLC 40 hosts the most launches and the early failures; KSC LC 39A flights begin around flight 25.
- VAFB SLC 4E carries no payload above about 9,600 kg; only CCAFS and KSC fly the heaviest loads.

EDA with Visualization: Orbit





- VLEO missions only appear late in the program (after about flight 60), which is when heavy, reused boosters are flying.
- ISS and SSO / LEO missions are mostly successful; GTO shows the most failures, consistent with its energy-hungry profile.

EDA with SQL

Question	Result
Distinct launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload for NASA (CRS)	45,596 kg
Mission outcomes	100 success, 1 failure
Max payload booster family	All 12 are F9 B5 boosters (B1048, B1049, B1051, B1056, B1058, B1060)
Drone-ship landing failures, 2015	Jan: B1012, Apr: B1015, both from CCAFS LC-40
Drone-ship success, payload 4,000 to 6,000 kg	B1022, B1026, B1021.2, B1031.2

Landing outcomes, 2010-06-04 to 2017-03-20: No attempt 10, Success (drone ship) 5, Failure (drone ship) 5, Success (ground pad) 3, Controlled (ocean) 3, Uncontrolled (ocean) 2, Failure (parachute) 2, Precluded 1.

Interactive Map with Folium

Launch site	Latitude	Longitude
CCAFS LC-40	28.5623	-80.5774
CCAFS SLC-40	28.5632	-80.5768
KSC LC-39A	28.5733	-80.6469
VAFB SLC-4E	34.6328	-120.6107

- Green / red clustered markers show landing success per site (56 launches mapped).
- Mouse-position tool used to read coastline, rail, highway and city coordinates; distances from the haversine formula (R = 6,373 km).

INSERT: Screenshots of your Folium map: (1) all four sites, (2) green/red marker clusters, (3) distance lines with km labels.

Measured: CCAFS SLC-40 is about **0.51 km** from the nearest coastline point (haversine, 28.56367, -80.57163).

Plotly Dash Dashboard

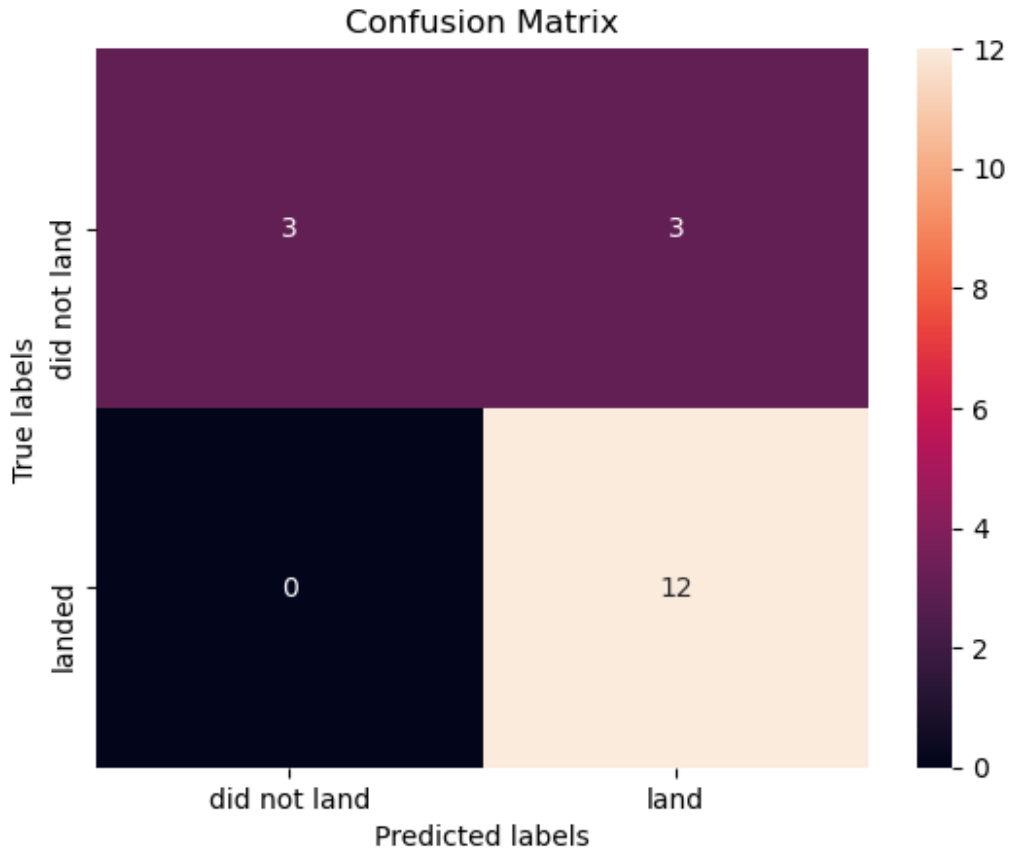
- Dropdown selects all sites or one site; range slider filters payload mass.
- **Pie chart** shows share of successful launches by site, or success vs failure for one site.
- **Scatter chart** plots payload vs Class, coloured by booster version.

INSERT: Screenshots of the Dash app (pie chart for all sites, the site with the highest success ratio, and the payload range scatter). No Dash file was among the uploads, so add app.py to the repo and answer: which site has the largest success share, which payload range has the highest success, and which booster version is best.

Predictive Analysis: Classification Results

Model	Best CV acc.	Test acc.
Logistic regression	81.9%	83.3%
SVM (rbf, C=1)	84.7%	77.8%
Decision tree (depth 2)	86.1%	83.3%
KNN (k=5, p=1)	84.7%	83.3%

- **Best model: decision tree**, highest cross-validation accuracy (86.1%).
- Logistic regression, tree and KNN tie at 83.3% on test (15 of 18 correct); SVM is lowest at 77.8%.
- The test set has only 18 launches, so one error moves accuracy by 5.6 points; the models are hard to separate.



Decision tree confusion matrix (test set)

Innovative Insights

15,600 kg
max payload: all Block 5 boosters

5 reuses
booster reuse count by flight 88

100 / 101
missions successful, per SQL

40%
Falcon 1 success rate, vs 100 of 101 missions now

- **Reuse and heavy lift go together:** the latest flights in the dataset (86 to 89) carry about 15,400 kg on Block 5 boosters with 2 to 5 prior reuses, and still land. Reuse is now routine even at maximum payload.
- **Learning curve:** early drone-ship attempts failed at the same pad twice in 2015, while by 2017 successes (5 drone ship + 3 ground pad) matched failures. Flight number acts as a proxy for operational maturity.
- **For Space Y:** a bid model should discount launch cost by landing probability, with flight experience, Block and payload as the strongest inputs.

Code and Repository

All work is in one GitHub repository: **https://github.com/Ali2187/spacex**

File	Purpose
jupyter-labs-spacex-api (data collection)	SpaceX REST API collection
jupyter-labs-webscraping	Wikipedia Falcon 9 scraping
labs-jupyter-spacex-data-wrangling	Class label, launch counts
jupyter-labs-eda-sql	10 SQL tasks in SQLite
edadataviz	seaborn / matplotlib EDA
lab-jupyter-launch-site-location	Folium maps and distances
app.py	Plotly Dash dashboard
SpaceX_Machine-Learning-Prediction	Four classifiers, GridSearchCV

Conclusion

- Landing success improves steadily with flight number, booster block and reuse experience.
- Heavy payloads (about 15,000 kg) fly almost only on Block 5 boosters from CCAFS and KSC; VAFB handles lighter polar missions.
- GTO is the hardest orbit to recover from; ISS, LEO and SSO missions land more often.
- A tuned decision tree predicts landing success with 86.1% cross-validated and 83.3% test accuracy, giving Space Y a data-driven estimate of whether a booster will be reused, and so the true launch price.
- **Next steps:** more launches (data ends Nov 2020), weather and booster-age features, and cross-validated model comparison given the small test set.

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